

Generative AI's Impact on Graduate Student Writing Productivity and Quality

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September 2024

Abstract

The rapid adoption of generative artificial intelligence (genAI) applications, such as ChatGPT, underscores their potential to significantly enhance writing. However, genAI efficacy varies among users and use cases, likely due to insufficient understanding of their operations and limitations. This study investigates the effect of genAI and targeted instruction on writing productivity and quality among graduate students on graded assignments. Employing a mixed-methods design, we compared students' performance on professional writing tasks conducted without and with genAI assistance. Comprehensive genAI instruction was provided prior to tool usage. Results showed genAI, with proper instruction, reduced writing time by 64.5% and improved writing quality from a B+ to an A. Benefits were observed across native English and ESL students, with slightly more pronounced benefits for ESL students, suggesting a leveling effect. This study highlights the importance of tailored educational strategies in leveraging genAI tools to maximize benefits and mitigate limitations.

1 Introduction

Generative artificial intelligence (genAI) applications, such as ChatGPT, demonstrate the ability to respond to human prompts on various subjects and tasks at near-human skill levels by representing patterns and logic from their training data^{1,2}. These capabilities are particularly relevant in educational and professional settings, where the demand for high-quality, efficient writing is paramount. One of the significant advantages of this technology is its accessibility; anyone with internet access and basic language skills can use genAI.

The ease of use and advanced capabilities of genAI have led to its widespread adoption. As of March 2024, 23% of US adults reported using ChatGPT³. A January 2024 study in Denmark found that nearly half of workers in roles where genAI could be relevant are utilizing this technology⁴. GenAI usage is also prevalent in higher education with 37% of students in the US and 53% in the UK using it as of February 2024^{5,6}.

Despite its widespread adoption, effectively utilizing this probabilistic technology can be challenging. GenAI's capabilities vary widely across tasks and it is difficult to determine what it can and cannot do, making it hard to predict its performance on any individual task⁷. Given genAI's profound potential, a critical topic for educators is how to teach about its effective use.

A primary use case for genAI is as a writing assistant given its ability to help users produce “semantically meaningful” and “grammatically correct” text⁸. Among one sample of college students who use ChatGPT for schoolwork, the top two applications are assisting with writing (69%) and research (67%) assignments⁵. In professional fields, strong research and writing skills are essential. Employers value these abilities, and educators aim to prepare students for the workforce by emphasizing them.

Investigating ways to effectively teach students to write with genAI is a current focus of many educational researchers.

1.1 GenAI in Professional Contexts

Early studies suggest genAI can enhance writing productivity and quality in professional contexts. Noy and Zhang found that ChatGPT with GPT-3.5 reduced average time spent by 40% and increased output quality by 18% in a set of simulated mid-level professional writing exercises. This was despite “almost everyone submitt[ing] lightly edited or unedited ChatGPT output”⁹ – far from a best practice. An in-depth study of consultants using ChatGPT with GPT-4 (a more capable model than GPT-3.5) on high-level professional writing and consulting tasks found that participants completed 12% more tasks with 40% higher quality results than a control group without genAI access⁷. These positive results were for an exercise designed to be within the assistance capabilities of ChatGPT. However, the same study found that on tasks outside ChatGPT’s capabilities, consultants were 19% less likely to produce correct solutions compared to those without genAI tools. Another study utilizing human annotators and GPT-4 classifications of U.S. workforce tasks demonstrated that genAI can significantly enhance workforce writing productivity by assisting with writing tasks¹⁰. These studies show the potential of genAI to improve writing productivity in professional settings but that genAI may not benefit all writing tasks equally.

1.2 GenAI in Educational Contexts

GenAI can also be a productive tool in academic settings. A classroom-based study involving English-as-a-second-language (ESL) students found that using genAI as a writing feedback tool boosted students’ academic writing quality¹¹. Additionally, these students’ perceptions of genAI as an assistive technology were quite positive. Another mixed-methods classroom study with ESL students found that students improved writing quality following a four-week genAI intervention¹². They also found that both students and teachers had improved perception of the benefits of ChatGPT in assisting with writing

education following this intervention. Similar productivity increases with genAI were found in a study of fiction composition among high-school students. Students writing short stories had reduced grammar and spelling mistakes and fewer plot holes when writing with genAI assistance as compared to a control exercise¹³. These studies begin to show the potential value of incorporating genAI into various academic writing tasks.

1.3 Challenges of genAI

However, despite these benefits of writing with genAI, studies also indicate that genAI does not always have a positive impact. In a consulting exercise specifically designed to expose potential errors in ChatGPTs analysis, consultants using genAI provided more incorrect solutions than those who did not use the technology⁷. This highlights the risk of uncritical reliance on genAI. Another study in a college writing class with genAI-assisted writing instruction provides a similar example. Here, most students declined to use the technology as they found the time required to review the often unreliable genAI suggestions was too great¹⁴. In another study, law students showed a mixture of improvements and declines in the quality of their written essays when using genAI assistance leading to a net neutral change in their grade¹⁵. The researchers attributed this to students lacking the skill to integrate the genAI output into their own writing. In a literature review of ChatGPT's role as a writing assistant, Imran and Almusharraf⁸ found that ChatGPT has a mixture of promising prospects but also significant challenges. They emphasized the need for further research to understand genAI's role in the writing process. These studies highlight the need to better understand the limits of genAI assistance in the writing process and the importance of instructing users on its effective and ineffective use.

This inconsistency in the impact of genAI can be attributed to several usage challenges inherent in the probabilistic and open-ended nature of this form of artificial intelligence. The biggest challenge is that genAI can produce output that looks correct but is inaccurate, a phenomenon referred to as a 'hallucination'^{7,16,17,18}. This could lead to users relying on inaccurate outputs and/or users spending

unproductive time fact checking genAI-created answers. One frustrating form of hallucination is the tendency of genAI to create fake research citations^{7,16,18}. GenAI also struggles with managing long documents and other lengthy contexts, which hinders its effectiveness with in-depth research and writing tasks if the user does not know how to manage genAI effectively^{19,20}. GenAI can also produce unclear, dull, and redundant text^{16,18}, with tendencies toward verbosity and unnatural language patterns²¹. These results again highlight the need for instruction on effective and ineffective use of genAI tools.

Another challenge of using genAI effectively is discerning which tasks it can perform well and which it cannot. There is no comprehensive instruction manual or complete mapping of genAI's capabilities⁷ and genAI systems typically do not inform users when tasks exceed their abilities. Tasks genAI is likely to struggle with include describing specialized subject knowledge, advanced logical reasoning and math, requests that require high levels of detail, and requests that require real-time access to multiple webpages^{19,22}. This can create a general uncertainty among users about how and when to use genAI^{7,14}. Overall, uncritical usage of genAI, such as asking ChatGPT for a written output and merely copying and pasting it, will likely limit the positive impact of the tool and may even hinder human creativity²⁴. Gathering user opinions and testing the effectiveness of genAI tools for various specific tasks in the writing process could help inform how to teach students about its capabilities.

Generative AI also does not help all users equally and may benefit weaker performers more than stronger performers. While Noy and Zhang⁹ found genAI significantly reduced the amount of time it took to generate a written product, this increase in quality was greater for participants who performed less well on the first writing task. In Dell'Acqua et al.'s⁷ study of consultant tasks, access to GPT-4 boosted the scores of lower performers by 43% while initially high performers improved only 17%. Choi and Schwarcz's¹⁵ study found a similar leveling effect in which initially weaker law students saw large improvements in their exam performance while initially stronger law students may have actually performed worse with genAI

assistance. Understanding whether genAI tools have different impacts across student demographics could help inform educational approaches.

1.4 Research Gap and the Present Study

The reviewed literature highlights both the major potential of improved productivity when using genAI tools and major challenges in how to use the tools effectively. This emphasizes the impact potential of effective educational instruction on the affordances and limitations of genAI tools for writing tasks.

Indeed, Chiu²³ found that many students hold optimistic views of the possibilities afforded by incorporating genAI into the classroom, while at the same time students recognized the need to build their own genAI literacy. Lack of genAI literacy and training is also a primary barrier to genAI adoption in the workplace⁴. Existing research on how to train this literacy, particularly in a college classroom, is sparse.

This study seeks to bridge this knowledge gap by measuring how classroom instruction can impact genAI's contribution to writing productivity and quality among graduate students on professional writing exercises. Crucially, our study explores how instruction on different sub-tasks of the writing process can impact productive usage and therefore writing performance. The study also considers productivity impacts for ESL and native English speaking students to investigate whether genAI has a leveling effect or may in fact increase inequality in writing performance. The study also seeks to measure students' perceptions about the usefulness of genAI in the research and writing process. The overall design is meant to help identify educational strategies that can better equip students in higher education for a professional landscape that is increasingly reliant on AI tools.

In this study, we employed a mixed-methods approach to examine the effects of genAI assistance on a graded writing assignment. Participants were a mix of native English speaking and ESL graduate students from a variety of academic disciplines. They were first assigned a professional memo writing task to be completed without the use of genAI. Students were then given classroom instruction on genAI system

development, operation, prompt engineering, failure modes, and demonstrations and exercises across research and writing sub-skills (i.e., style, flow and concision; organization; summarization; external research; research analysis; and policy development). Students were then provided with an isomorphic professional writing exercise for which they could use genAI assistance if they chose (100% of students used genAI assistance). By comparing performance across these conditions, we aimed to determine the impact of genAI on students' efficiency and writing quality.

The research questions we investigate are:

1. How does instruction about genAI and use of genAI as a writing assistant impact research and writing productivity and quality?
2. Does this instruction and genAI assistance have different impact across research and writing sub-tasks (e.g., summarization, organization, analysis)?
3. Does the impact of instruction and genAI assistance differ across writers with different English language backgrounds?
4. What are students' perceptions of genAI?

2 Methods

2.1 Participants

Participants were 28 graduate students enrolled in the graduate course toward a degree program during the spring 2024 term. The course was a 7-week graduate elective class at Carnegie Mellon University. In accordance with our exempt IRB protocol (approved by the Carnegie Mellon University Institutional Review Board, STUDY2016_00000148), potential participants were notified verbally and via their syllabus that educational research was happening in their course and that they had a right to opt out of

having their course data used for research purposes. One student opted out of the study, resulting in a final sample size of 27 participants.

Participants were from a variety of academic disciplines including information systems and public policy (63%), business (25.9%), computer science (7.4%), and humanities (3.7%). One participant was enrolled in a doctoral program and all remaining participants were in Master's programs. Fifteen participants (55.6%) self-reported being English as a Second Language (ESL) student while 12 participants (44.4%) were native English speakers.

Eleven participants (40.7%) self-reported having an extensive technical background in computer science or programming. Ten participants self-reported having no technical background and six reported having a little background. Twenty-three (85.2%) participants reported having prior experience using ChatGPT or other AI to help with their coursework.

2.2 Survey Measures

Participants took a pre-class survey to gather demographic information and their experience with AI.

They self-reported whether or not they are a native English speaker, their graduate program enrollment, technical background in computer science or programming (none, a little, extensive), whether or not they had previously used ChatGPT or other AI to help with coursework (yes, no), and how they self-evaluate their professional English writing competency compared to other graduate students at CMU (1 = worse than almost all graduate students at CMU, 5 = Better than almost all graduate students at CMU).

Additional questions were included that were not related to the research study but rather provided the course instructor with an overview of his students and their learning goals.

Participants took a post survey to determine how students used the genAI tool in the course and their views about the tool. They self-reported on which research and writing sub-tasks they used AI

(summarization, web research, policy development, analysis, writing, or none of the above), how useful the AI tool was in each of the above tasks (1 = not useful, 4 = very useful), how much they believe their research and writing quality improved from the first to the third assignment (1 = did not improve, 5 = significantly improved), how much they believe their research and writing speed improved from the first to the third assignment (1 = did not improve, 5 = significantly improved), how much they attribute any improvement in writing quality and/or speed to the use of AI versus experience gained in completing similar assignments (1 = 100% from completing similar assignments, 5 = 50/50 improvement from experience with a similar assignment and benefits from using AI, 10 = 100% from using the AI tool), and how much they value the benefits of AI tools to their writing quality versus speed (1 = 100% value the quality, 5 = equally value quality/speed improvement, 10 = 100% value the speed). Participants self-evaluated their professional English writing competency using the same question from the pre survey. The survey also provided participants the opportunity to provide open-ended responses about how AI most impacted their research and writing. Additional questions were asked to provide the instructor an overview of participants' opinions about the course and student attitudes.

2.3 Writing Assignments and Grading Rubric

A sequence of professional memo writing exercises was designed to consistently involve the same five core professional research and writing sub-tasks of summarization of a news article, web research to find survey results, analysis and synthesis of different survey results, policy development, and critical analysis. In addition to requiring students to use these same skills, each assignment followed the same structure. The assignment prompt was written as if from a different stakeholder (e.g., the CEO), highlights an observation about AI based on a research study or headline, shares two results from the source, requests a deliverable that summarizes the cited source, calls on the student to do additional research to assess the representativeness of the results, requires them to explain the risks and benefits of the generative AI in the particular context, and finally to present a well-argued policy for the specific work context (see Supplementary Material 1 for a sample assignment).

Assignment version A was in the context of a university. Version B was in the context of an outdoor goods business. Version C was in the context of a newsroom. All students completed all three assignment versions, but in a random order, meaning each version of the assignment appears in each of the three instructional conditions (without genAI, with extensive use of genAI, and with genAI assistance).

The three assignment deliverables were evaluated using the same grading rubric (see Supplementary Material 2). All rubric criteria were scored on a scale of 1 (insufficient), 2 (sufficient), or 3 (proficient). Quality of summarization was a single criterion. External survey findings involved two criteria of quality of external survey research as well as integration and analysis of survey results. Policy recommendation involved three criteria of articulation of their policy, how logical and comprehensive their policy was, and how well they argued and supported their policy with analysis. Writing style had two criteria of style, flow, and concision as well as organization. The maximum grade on each memo assignment was 24 points.

2.4 Procedure

All participants completed three writing assignments in the order of no genAI, full genAI, and genAI-assisted. The specific topic of each writing assignment was counterbalanced across conditions. The full genAI assignment is not a data source for this study but was rather an instructional tool used to help teach students about the affordances and limitations of genAI.

Participants completed the pre survey immediately before the start of the course. Participants were informed about the educational research via their course syllabus and verbally during the first class session. Participants were randomly assigned writing assignment version A, B, or C to complete the first week of classes. This first assignment represented the no genAI condition. To increase compliance, the instructor presented this assignment within the context of the educational research study and told

participants to complete their writing assignment as they typically would without any use of genAI during the research or writing process. This was a take-home assignment and participants could spend as much or as little time completing it as they chose. They were told to track and calculate the amount of time they were actively engaged in completing the assignment, excluding any breaks. They were encouraged to use a device like their phone's stopwatch to track this time and were instructed to estimate to the best of their ability if they forgot to start or stop the timer. They were also told that the amount of time they spent on the assignment would not impact their grade for the assignment.

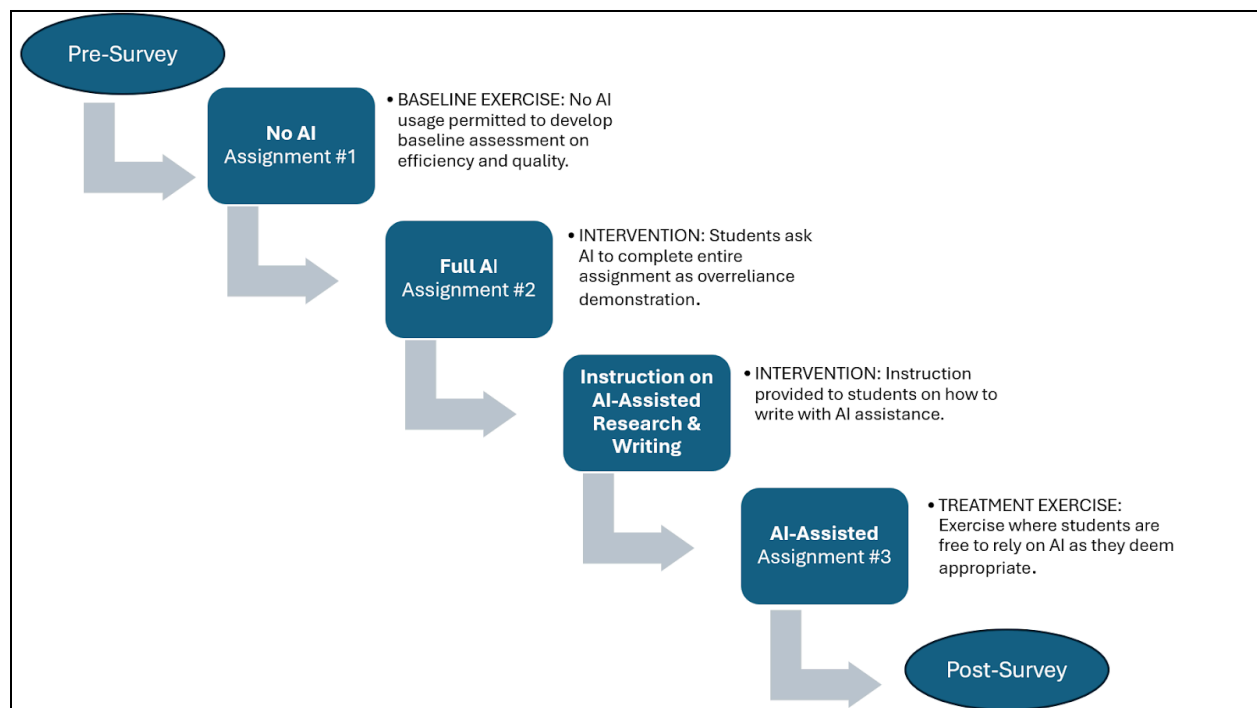
The second assignment was the full genAI condition. This was an in-class assignment during the second week of the course and participants were only permitted 30 minutes to complete it. This condition is not part of the comparison of this study as it primarily served as an instructional tool. Students were told to rely on the AI as much as possible for the assignment, such as submitting the entire prompt into the generative AI system to see how capable the AI is of completing the assignment on its own. The goal of this exercise was to teach students the challenges of over reliance on genAI and to demonstrate the tool's strengths and weaknesses. This assignment was not a data source for this study.

After the first assignment but before the third, the course instructor provided several hours of take-home reading and in-class instruction on how to effectively co-author with genAI. The goal of this instruction was to teach students techniques to appropriately use gen AI during the research and writing process. However, this was not only focused on skills to write with AI. The learning path also covered the basics of how genAI large language models are developed. As it is challenging to understand what genAI systems can and cannot do, a basic understanding of how they work is crucial to develop student inference into what the models can help with and their limitations. This basic understanding is also important to build student resilience to keep up with advancements in genAI capabilities. These learning units spanned two to three weeks of the course and are described in Supplementary Material 3.

After this instruction, students were given their third writing assignment. Students could choose whether to complete this assignment with genAI assistance or independently without genAI. Every student chose to use genAI as an assistant and so this third assignment represents our genAI-assisted condition. Students were able to choose their own genAI application with all choosing either ChatGPT or Microsoft Copilot. Students were again told to track and calculate the amount of time they were actively engaged in completing the assignment, excluding any breaks.figure

During the last week of the course and before receiving their grades for the third writing assignment, students completed the post survey to better understand the ways in which students used genAI in their third assignment and their views about the utility of the tool. Figure 1 shows the overall method and sequence of events.

Figure 1: Sequence of events in the present study.



3 Results

Two versions of a memo writing assignment were evaluated for this study. Both writing assignments were structured the same and were graded using the same rubric (see Method and Supplementary Material 2).

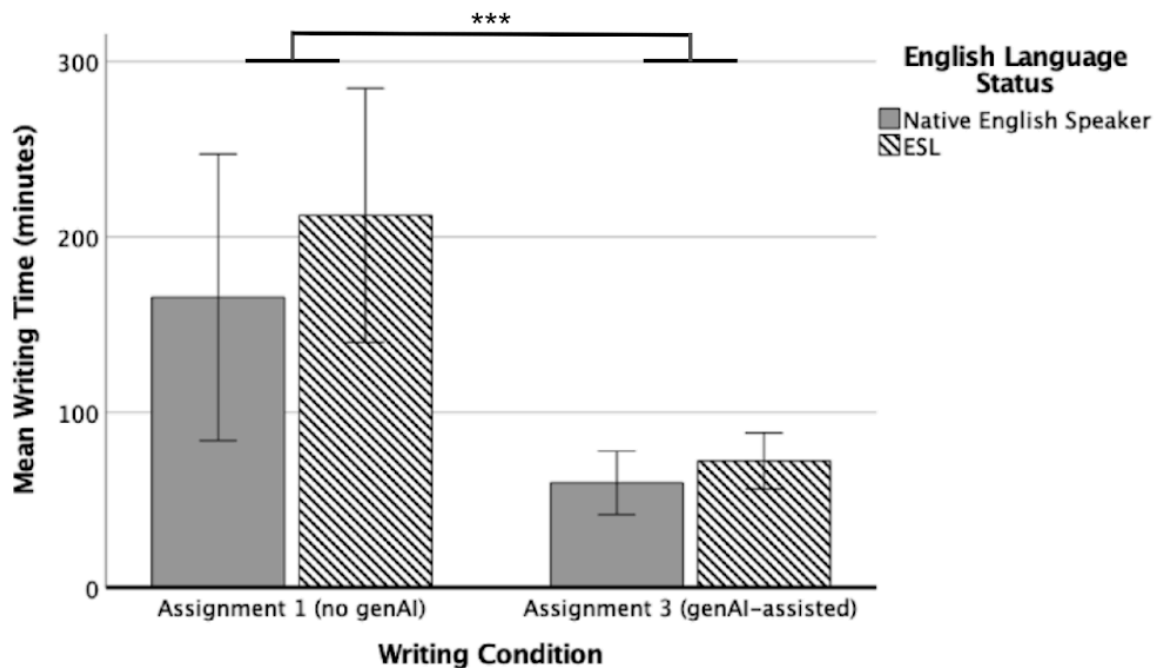
The first was completed without the use of genAI and the latter was completed with optional genAI assistance. Every student reported using genAI for their genAI assisted memo writing assignment. Out of the 29 students in the study, many students reported using the tool for writing sub-tasks, specifically summarization (23 students), writing (22 students), and/or web research (22 students). Additionally, 18 students reported using the tool for policy development (e.g., articulating the workplace policy in the assignment scenario) and 18 students used it for analysis (e.g., of survey results, policies, arguments). On average, students used the genAI tool for assistance in 3.8 of these 5 sub-task areas.

3.1 Effect on Research and Writing Speed

Students were asked to keep track of how long they were actively engaged in completing their memo writing assignments (both research and writing) and to submit this time along with their deliverable. A 2x2 mixed-model ANOVA tested for the difference in this writing time over writing condition (no genAI, genAI-assisted) and English language status (native English speaker, ESL). There was a significant main effect of writing condition on writing time ($f(1,23) = 23.15, p < .001, \eta_p^2 = .50$) showing that students spent significantly less time writing their memo when they used genAI assistance. On average, genAI assistance reduced the time spent completing their policy memo assignment by 64.5%. There was no main effect of English language status ($f(1,23) = 1.09, p = .31, \eta_p^2 = .05$) meaning native English speakers and ESL students spent comparable time preparing their memo writing assignments. There was no interaction between writing condition and English language status on writing time ($f(1,23) = .45, p = .51, \eta_p^2 = .02$) meaning, statistically, the amount of time saved writing with genAI assistance did not differ

between native English speakers and ESL students (see Figure 2). However, the trend in the data suggests that ESL students showed slightly greater improvements in productivity, though this difference was not statistically significant. These results suggest that genAI assistance can improve the efficiency of the research and writing process for students with different English language backgrounds.

Figure 2: Effect of writing condition and English language status on time spent completing the writing assignment. Bars represent group means. Error bars represent 95% confidence intervals. *** $p < .001$.

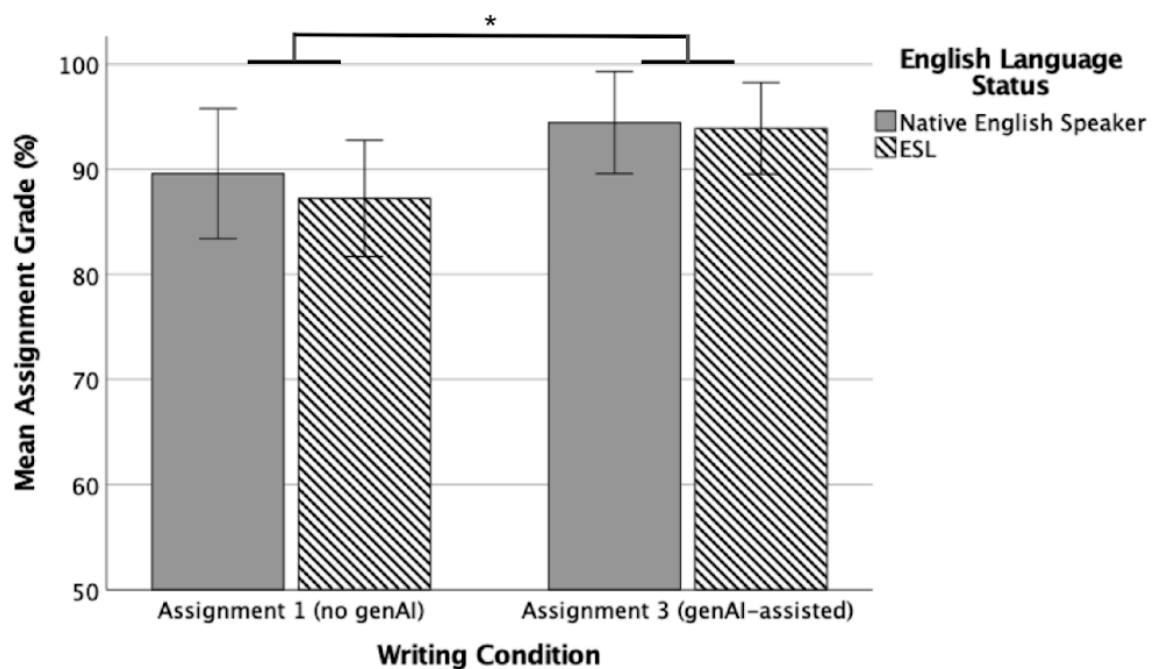


3.2 Effect on Writing Performance

Writing performance was defined by students' earned grades on a standardized grading rubric (see Supplementary Material 2). A 2x2 mixed-model ANOVA tested for the difference in writing performance across writing conditions (no genAI, genAI-assisted) and English language status (native English speaker, ESL). There was a significant main effect of writing condition ($f(1,25) = 4.74, p = .04, \eta_p^2 = .16$) indicating that students earned higher grades on the assignment prepared with genAI assistance. On average, students' grades change from a B+ ($M = 88.3\%, SD = 10.3\%$) without genAI to an A ($M =$

94.1%, $SD = 8.0\%$) with genAI assistance. There was no main effect of English language status ($f(1,25) = .35, p = .56, \eta_p^2 = .01$) meaning ESL students and native English speakers earned equivalent assignment grades. There was no interaction of writing condition and English language status ($f(1,25) = .12, p = .74, \eta_p^2 = .01$) meaning the amount of change in assignment grades did not differ between native English speakers and ESL students (see Figure 3). However, the trend in the data suggests that ESL students showed slightly greater improvements in performance, though this difference was not statistically significant. These results suggest that genAI assistance can equally improve the quality of the written deliverable for students with different English language backgrounds.

Figure 3: Effect of writing condition and English language status on writing assignment grade. Bars represent group means. Error bars represent 95% confidence intervals. * $p < .05$.



The 8 sub-tasks that composed the overall writing assignment grade (see Appendix B) were also analyzed to see if there was significant change from the first (no genAI) to last (genAI-assisted) writing assignment. See Table 1 for a summary of these results.

Sub-Tasks	Asgmt 1 (no genAI)	Asgmt 3 (genAI-assisted)	Degree of improvement	Statistical Result
Organization	2.70 (.47)	2.89 (.32)	.19 point or 6.33%	$f(1,26) = 3.96, p = .06, \eta_p^2 = .13$
Style, flow, and concision	2.52 (.58)	2.74 (.45)	.22 points or 7.33%	$f(1,26) = 3.25, p = .08, \eta_p^2 = .11$
External survey research	2.67 (.78)	2.89 (.42)	.22 points or 7.33%	$f(1,26) = 2.36, p = .14, \eta_p^2 = .08$
Summarization	2.78 (.42)	2.93 (.27)	.15 points or 5%	$f(1,26) = 2.08, p = .16, \eta_p^2 = .07$
External survey analysis	2.56 (.75)	2.78 (.51)	.22 points or 7.33%	$f(1,26) = 1.41, p = .25, \eta_p^2 = .05$
Policy argument	2.67 (.48)	2.81 (.48)	.14 points or 4.67%	$f(1,26) = 1.15, p = .29, \eta_p^2 = .04$
Policy articulation	2.70 (.61)	2.81 (.40)	.1 points or 3.33%	$f(1,26) = .68, p = .42, \eta_p^2 = .03$
Policy logic and holistics	2.63 (.57)	2.67 (.48)	.04 points or .16%	$f(1,26) = .06, p = .81, \eta_p^2 = .00$

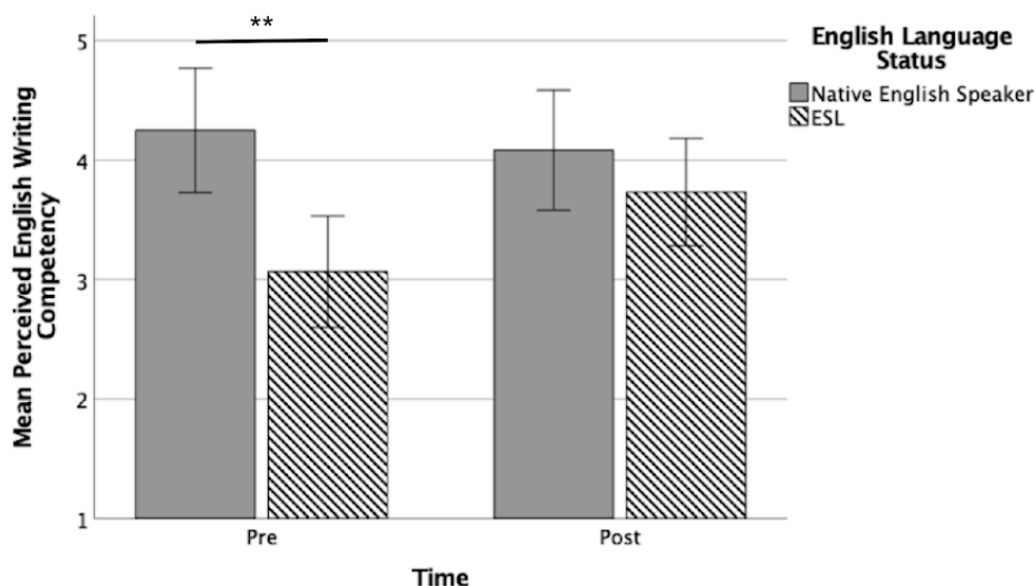
Table 1: Summary of performance change across sub-tasks. Rows represent sub-tasks that were evaluated with the grading rubric. The mean assignment grades (points earned out of 3 possible) in the two writing conditions are presented in the second and third columns with the standard deviation in parentheses. The fourth column displays the amount of change in points earned and grade percentage between the two iterations of the assignment. The repeated-measures ANOVAs testing for a difference between writing conditions are reported in the last column. Rows are ordered from largest to smallest statistical effect based on effect size. The reported *p*-values are uncorrected for multiple comparisons.

3.3 Effect on Perceived Writing Competency

In addition to measuring the impact of genAI on writing efficiency and quality, we were interested in the impact on students' perceived English writing competency and whether this effect would differ between students based on their English language status. Students self-reported their English writing competency at the beginning and the end of the semester. A 2x2 mixed-model ANOVA tested for the difference in

perceived writing competency over time (pre, post) and English language status (native English speaker, ESL). There was a significant effect of English language status on perceived writing competency ($f(1,25) = 8.70, p = .007, \eta_p^2 = .26$) with ESL students reporting lower perceived competency than native English speakers. There was no main effect of time ($f(1,25) = 1.44, p = .24, \eta_p^2 = .05$) indicating that students' perceived writing competency did not significantly change from the beginning to the end of the course. There was, however, a marginally significant interaction between English language status and time ($f(1,25) = 3.99, p = .06, \eta_p^2 = .14$, see Figure 4). Follow-up pairwise testing found that coming into the course, ESL students self-evaluated their English writing competency significantly lower than did native English speakers ($t(25) = 3.49, p = .002, d = 1.35$). By the end of the semester, this difference had disappeared ($t(25) = 1.07, p = .30, d = .41$) with ESL students reporting equivalent perceived writing competency to their native English speaking peers. Additionally, ESL students showed a marginally significant increase in their perceived writing competency over time ($t(14) = -2.09, p = .06, d = .54$) whereas native English speaking students did not change in their perceived competency ($t(11) = .69, p = .50, d = .20$).

Figure 4: Effect of time and English language status on students' perceived English writing competency. Bars represent group means. Error bars represent 95% confidence intervals. $**p < .01$.



3.4 Student Attitudes and Perceptions

To help contextualize change in students' performance on their third writing assignment (genAI-assisted), students rated how useful they found the AI tool to be in the various assignment sub-tasks. Students found the genAI's assistance to be the most useful for summarization ($M = 3.52$ (max = 4), $SD = .75$) and assistance in writing ($M = 3.50$, $SD = .71$). The tool's usefulness for the sub-tasks of web research ($M = 3.08$, $SD = .81$), policy development ($M = 3.04$, $SD = .82$), and analysis ($M = 2.87$, $SD = .92$) was a bit lower. ESL students found the genAI tool to be significantly more useful in the writing process than did native English speaking students ($t(24) = -2.09$, $p = .047$).

Students were asked how much they thought their research and writing speed improved on the genAI-assisted memo assignment. They reported feeling their speed had improved considerably ($M = 4.44$ (max = 5), $SD = 1.01$). Native English speaking students reported feeling more significant improvement in their speed than did ESL students ($t(15.92) = 2.60$, $p = .02$). Students attributed their

improvement in writing productivity and quality more to GenAI (66.7% weight) than repeated practice with assignments.

Students were also asked how much they thought their research and writing quality improved on the genAI-assisted memo assignment. They reported feeling like their quality had somewhat improved ($M = 3.74$ (max = 5), $SD = .76$). This sense of improvement in quality was slightly higher in ESL than native English speaking students though this failed to reach statistical significance ($t(25) = -.96, p = .35$). This perception of quality significantly correlated with the change in total assignment grade from assignment 1 (no genAI) to assignment 3 (genAI-assisted) ($r(27) = .45, p = .02, R^2 = .20$) such that students who perceived a greater improvement in their writing quality showed a larger increase in their grade. Note, this self-assessment of their quality was measured prior to students receiving their grade on the genAI-assisted writing assignment.

As a final means of capturing students' perceptions of working with the genAI tool, they were asked how the AI impacted their research and writing process. Twenty-one students provided qualitative responses to this open-ended question which were coded using thematic analysis. Eight students reported improved writing speed (e.g., fewer drafts). Seven students reported improved research process (e.g., finding articles, speed of process). Six students reported improved writing quality (e.g., grammar and accuracy). Five students reported on the tool's ability to support summarizing and/or consolidating information. Three students reported on the tool's benefit as a writing assistant (e.g., refining the student's draft). Despite the question being worded neutrally, all students reported positive impacts and no students reported a negative impact on their research or writing process.

4 Discussion

In this study, we examine how a method of genAI instruction, and genAI usage on a research and writing assignment following this instruction, impacted student writing performance and efficiency. The classroom instruction covered how genAI systems are developed and operate, prompt engineering, avoidance of genAI failure modes, and in-class demonstrations and exercises across genAI writing sub-tasks (see Method for additional details).

4.1 Improvements in Writing Speed and Quality

We find that genAI assistance following this instruction significantly improved writing productivity, reducing the time spent on research and writing tasks by 64.5%, from a mean of 191.8 minutes ($SD = 130.3$ minutes) on the no-genAI assignment to a mean of 68.1 minutes ($SD = 28.4$ minutes) on the genAI-assisted assignment. GenAI use also significantly boosted quality, as average students' grades improved from a B+ ($M = 88.3\%$, $SD = 10.3\%$) without genAI to an A ($M = 94.1\%$, $SD = 8.0\%$) with genAI assistance. In a context where grades are highly compressed, this shift represents a meaningful improvement from strong to even stronger performance.

Students valued the improvement in their writing speed more than the improvement in the quality of their writing. When asked "How do you value the benefits of AI tools to your research/writing quality versus speed?" with response anchors of 1 = 100% value quality, 5 = equally value quality and speed, and 10 = 100% value speed, the average was 6.7. When asked how much they thought their research and writing speed improved on the genAI-assisted memo assignment, they reported feeling like their speed had considerably improved ($M = 4.44$ (max = 5), $SD = 1.01$). When asked the same question about research and writing quality, they reported a weaker sense of improvement ($M = 3.74$ (max = 5), $SD = .76$). When asked "How much do you attribute the improvement in your writing quality and/or speed to the use of AI

versus your gained experience in completing similar assignments?” on average, students attributed the change more to genAI than repeated practice (67% due to genAI).

Our findings align with Noy and Zhang’s⁹ study. They reported that genAI significantly reduced writing time (40% decrease in time) along with an improvement in writing quality (18% increase in quality). Our results extend Noy and Zhang’s findings from college-educated, working professionals’ performance on a mock writing task to graduate students’ performance on a graded academic writing assignment designed to replicate writing in a professional setting. Additionally, we found that both weaker and stronger writers improved in speed and quality whereas Noy and Zhang found quality improvements predominantly for initially weaker writers.

4.2 Impact on ESL Students

The present study also explored genAI's impact on ESL students compared to native English speakers. While not statistically significant, likely due to the small sample size, the observed student data show that ESL students improved more in both productivity and quality compared to native English speaking students. In productivity, ESL students spent on average 212.36 minutes ($SD = 116.75$ min) preparing their first assignment without genAI, while native English speaking students spent on average 165.64 minutes ($SD = 147.32$ min). With genAI assistance, ESL students closed the time to completion gap, averaging only 10 minutes longer to complete the assignment compared to native English speakers, down from an average 47 minute gap on the first assignment. A similar leveling effect held for quality. The observed data show ESL students had an 85% average grade on the no-genAI assignment compared to 87% for native English speaking students. In the final assignment with genAI assistance, cohorts earned equivalent grades ($M_{\text{Native}} = 94.4\%$, $M_{\text{ESL}} = 93.9\%$). However, this finding was also statistically insignificant.

Our qualitative survey results agree with our quantitative data. Both findings suggest a leveling effect of genAI for ESL students. On the survey taken at the beginning of the course, ESL students self-evaluated their English writing competency significantly lower than native English speakers ($p = .002$). By the end of the course, this difference had disappeared ($p = .30$), with ESL students reporting equivalent perceived writing competency to their native English speaking peers. This shows that instruction on how to use genAI and practice using it can improve equity in educational outcomes by boosting writing efficiency, quality, and confidence among ESL students. Importantly, students with stronger self-assessed writing competency were not negatively impacted by using the genAI tools. They merely did not have the same confidence growth as their ESL peers. ESL students additionally found the genAI tool to be significantly more useful than did native English speaking students ($p = .047$). Our findings align with Mahapatra's¹¹ study of an ESL classroom which found that using genAI as a writing feedback tool boosts ESL students' academic writing quality. Our study extends these findings by showing that ESL students can benefit from genAI as a general research and writing assistant, beyond its narrow use as a feedback tool.

While the quantitative difference between ESL and native English speakers were not statistically significant, the observed findings mirror previous research on the leveling effect of genAI for weaker performers^{7, 9, 15}. This study provides further evidence that genAI can be an equality leveler for students writing in a second language even in the context of a highly competitive graduate program. It also adds to prior scholarship showing that workers with less prior experience could have the most to gain from genAI²⁵. This leveling effect warrants continued research as it can help lower performers catch up to top performers, reducing inequality in educational and workplace performance.

4.3 Impact on Research and Writing Sub-Tasks

The present study additionally aimed to evaluate the impact of genAI on sub-tasks within the research and writing process. This evaluation could help identify where genAI could be the most helpful or determine what tasks are beyond genAI's current capabilities, known as the 'technological frontier'⁷. Table 1 shows

the sub-tasks and the degree of grade improvement from completing the assignment without genAI to completing the assignment with genAI assistance.

GenAI improved students' grades across all but one sub-task, with the most sizable gains in organization and style, flow, and concision. Given the small sample size, these changes were not statistically significant, but the effect sizes were large ($\eta_p^2 = .13$ and $.11$, respectively). Medium effect sizes were found for the sub-tasks of summarization and finding and analyzing external research. These results suggest that these tasks are squarely within genAI's technological capabilities. Indeed, students reported finding the tool to be particularly useful for summarizing and writing assistance. These results also align with the pedagogical approach used, which taught students how to use genAI for these organizational tasks and as a feedback tool to help students rephrase challenging text. Students were additionally taught how to use genAI to find relevant external research on the internet and how to compare and contrast multiple survey results.

Small effect sizes were found for the sub-tasks of creating and articulating a policy argument. This could indicate that these tasks are outside of the current abilities of the genAI tools. This is despite prior class instruction on how to use the tool to develop arguments and counterarguments and to help students craft a policy. Lastly, policy logic showed no quality improvement when using genAI. Students were asked to craft policies for various contexts (i.e., for a University, newspaper, or business) on the permitted usage of genAI tools. Fewer students used genAI for policy development than other sub-tasks, perhaps another cause of a small effect size. This is a complex task that must account for the benefits of genAI, the ethical challenges, and how to balance these with any unintended consequences of a policy within the assigned context. This finding aligns with Dell'Acqua's research, which indicates that genAI's effectiveness depends on whether tasks fall within the "jagged technological frontier" of the tool's capabilities⁷.

4.4 Student Perceptions

As a final means of capturing students' experience working with the genAI tool, they were asked in a post-treatment survey as a series of closed- and open-ended questions about how genAI impacted their research and writing process and perception of genAI.

Students reported that the genAI tool helped with research and writing speed, quality and accuracy of their writing, and ability to summarize information. These student attitudes about the tool matched the changes in writing quality and reported speed that we observed.

Demonstrative student quotes:

- *“The largest impact was on summarization, saving time finding sources, and generating ideas from a draft or outline I created.”*
- *“It helped me organize my thoughts/create an outline and helped me speed up my writing process.”*
- *“Helped me go faster and I learned ways to improve the output of my AI answer from ChatGPT along with mixing and matching tools for their strengths.”*
- *“AI had allowed me to find articles and get an aggregate or median opinion from their contents and return a writing sample of that combined opinion.”*
- *“I mostly used it for summarizing research articles and ensuring that my writing and grammar were accurate.”*

Students were also enthusiastic about using genAI and the importance of genAI instruction. All students reported they intended to use the tool in future writing tasks. Also, a majority of respondents (16 students, 59.26%) believed that “a class that teaches AI assisted productivity, such as AI-enabled research and writing, should be required to receive a graduate degree from CMU.” This shows that students recognize the benefits of genAI instruction.

4.5 Implications and Recommendations

Our results are further evidence that teaching students how to appropriately leverage genAI for writing assistance is becoming an essential job for educators. Higher education should consider incorporating genAI instruction into curriculum across undergraduate and graduate programs. Moreover, given lack of genAI training may be a primary barrier to adoption for professionals, the present instructional method may be useful both for educational institutions and for workplace training⁴. The instruction covers how genAI operates and is developed, including instruction on prompt engineering, and describes genAI assistance techniques across the research and writing process (see Supplementary Material 3). Given the observed benefits, other colleges could consider adopting this curriculum template to enhance writing proficiency and prepare students for future AI-integrated work environments. Given the swift advancement in genAI, ongoing research will be needed to tune the instructional path to keep up with the latest capabilities of genAI.

If the present findings on writing productivity enhancement apply to the workplace—which Noy and Zhang’s⁹ results would suggest—genAI could significantly boost productivity in the economy. This implies that in the coming years, employers will be motivated to demand that employees know how to write with genAI assistance, putting pressure on educators to incorporate genAI into their curriculum to prepare students for an AI-enabled workforce.

This study also provides preliminary evidence of a leveling effect, with greater improvement for ESL students for writing speed and quality. Choi and Schwarcz’s¹⁵ found a similar leveling effect in which initially weaker law students saw larger improvements in their exam performance than initially stronger students. The potential ability of genAI to help lower-performing students catch up to their peers could help level educational outcomes. Further research is needed to test this impact on ESL versus native English writers.

The study is also a research design template for other genAI education and workforce use cases such as coding, legal analysis, medical treatment, artistic design, and myriad others^{15,26,27,28}. GenAI has shown promise across these domains and there is an urgent need to understand its impact and to develop educational strategies to equip students and those in the workforce with skills to maximize genAIs benefits while managing its challenges and ethical quandaries. Universities will likely need both broad and deep educational strategies for genAI literacy. Broad strategies can cover aspects such as prompt engineering and writing, which are complemented by deep strategies tailored to specific academic areas.

Future research is needed to determine the appropriate timing for incorporating genAI assistance into curricula. Similar to the delayed introduction of calculators until middle or high school, students should first learn intermediate to advanced writing skills without genAI. This approach should ensure that they can maximize genAI's potential. In other words, students must become proficient writers themselves before they can effectively "manage" and utilize genAI capabilities.

4.6 Limitations

The present study emphasized ecological validity by testing the impact of leveraging genAI in the research and writing process of a real graduate school classroom, with real students, using a pedagogically integrated method to create an intervention and comparison assignment. Prioritizing this ecological validity came at a cost of internal validity. While the present study did involve a comparison condition in which students completed their writing assignment without genAI assistance, due to logistical and pedagogical constraints we were not able to counterbalance the order of writing conditions. While we found significant effects on student performance (both writing speed and quality), we cannot definitively attribute that impact to genAI use alone. It is possible that other factors could have (partially) contributed to improvements in performance, such as repeated practice with the assignment, though when asked, students attributed their improvement more to the genAI assistance than practice. Future studies

could consider replicating in a controlled lab setting in which the order of conditions can be randomized to strengthen the causal inference.

Additionally, because this was a real classroom, the sample of the present study was constrained by the type and number of students enrolled. This resulted in a self-selected group of students interested in taking an elective course on AI, 85.2% of whom reported having used genAI to help with their previous course work. The sample size was 28 students. This was a limit on the power of this study to detect all of the possible effects of genAI assistance and disaggregated analyses testing whether there are differential effects for students with different English language backgrounds. For instance, while not statistically significant, our ESL students took longer to complete their first writing assignments and scored lower than native English speakers. Both speed and quality were much more similar when students used genAI assistance. This pattern and the other results of the present study are suggestive that writing with genAI assistance could serve as an equity tool to support all students in their research and writing process.

Another limitation is that the grading rubric was implemented by the professor, who was not blinded to the condition, which could impact the validity of the quality assessments. However, this is mitigated by a detailed grading rubric.

4.7 Conclusion

Our findings reveal that genAI use, following appropriate instruction, led to notable improvements in both productivity and quality of research and writing on a graded academic assignment. Importantly, these benefits accrued for both native English speakers and ESL students, with the observed data suggesting leveling effects that help ESL students catch up to initially higher-performing, faster peers. Moreover, these results suggest that genAI can have considerable impact on various sub-tasks across the research and writing process (e.g., summarization) while not impacting others (e.g., policy articulation). This suggests that some tasks are either beyond the current capabilities of genAI or that different pedagogical

approaches may be required for students to effectively leverage the tools for these purposes. Overall, these findings suggest that genAI has strong potential to help writers and that effective training can equip students for an AI-infused professional workplace. As generative AI is not likely to disappear from higher education, teaching students how to appropriately leverage the tool in their work is becoming an essential job for educators. The present study both provides instructional ideas of how to do this and how this is likely to impact student writing.

Study Administration and Acknowledgments

Research Ethics Statement

The study reported in this manuscript was approved by the Carnegie Mellon University Institutional Review Board (STUDY2016_00000148). This study complied with all relevant ethical guidelines.

Data Availability

The data are not publicly available due to the IRB protocol used to collect these data not permitting this. Aggregate data are available upon request from the corresponding author [ACP].

Acknowledgements:

The authors would like to acknowledge the amazing help of the Eberly Center and their staff! In particular, Jacqui Pincus who consulted on the instructional and research design for this project, providing pedagogical expertise in designing the assignment rubric. We also thank Provost James Garrett who provided internal funding to support this educational research project.

Competing Interests Statement:

A.C.P. and H.C. have no competing interests to declare.

J.U. has a potentially competing interest to declare. In addition to being an adjunct faculty at Carnegie Mellon, he is employed by Microsoft as the Senior Director of Strategy and Innovation in the office of the Chief Technology Officer.

Use of Generative AI in Authoring:

In the early stages of preparing this manuscript, J.U. utilized a genAI tool to aid in rephrasing certain sections for clarity and conciseness. The tool also served as a feedback mechanism for the overall written content. GenAI did not contribute to the conceptualization, design, data collection, analysis, or interpretation of the study, and thus, is not listed as an author.

References

1. Bubeck, S., Chandrasekaran, V., Eldan, R., Gehrke, J., Horvitz, E., Kamar, E., Lee, P., Lee, Y. T., Li, Y., Lundberg, S., Nori, H., Palangi, H., Ribeiro, M. T., & Zhang, Y. Sparks of Artificial General Intelligence: Early experiments with GPT-4, *arXiv*, arXiv:2303.12712 (2023).
<http://arxiv.org/abs/2303.12712>
2. OpenAI. *Introducing ChatGPT*. <https://openai.com/index/chatgpt/> (2022, November 30).
3. McClain, C. Americans' use of ChatGPT is ticking up, but few trust its election information. *Pew Research Center* (2024, March 26).
<https://www.pewresearch.org/short-reads/2024/03/26/americans-use-of-chatgpt-is-ticking-up-but-few-trust-its-election-information/>
4. Humlum, A., & Vestergaard, E. The Adoption of ChatGPT. *U. Chicago, BFI*, Working Paper No. 2024-50 (2024). <https://doi.org/10.2139/ssrn.4807516>
5. *4 in 10 College Students Are Using ChatGPT On Assignments*. *Intelligent*. (2024, February 27).
<https://www.intelligent.com/4-in-10-college-students-are-using-chatgpt-on-assignments/>

6. Freeman, J. Provide or punish? Students' views on generative AI in higher education. *HEPI*, Policy Note 51 (2024, February 1).
<https://www.hepi.ac.uk/2024/02/01/provide-or-punish-students-views-on-generative-ai-in-higher-education/>
7. Dell'Acqua, F., McFowland, E., Mollick, E. R., Lifshitz-Assaf, H., Kellogg, K., Rajendran, S., Kraye, L., Candelon, F., & Lakhani, K. R. Navigating the jagged technological frontier: Field experimental evidence of the effects of AI on knowledge worker productivity and quality. *Harvard Bus. School TOM*, Working Paper Number 24-013 (2023).
<https://doi.org/10.2139/ssrn.4573321>
8. Imran, M., & Almusharraf, N. Analyzing the role of ChatGPT as a writing assistant at higher education level: A systematic review of the literature. *Edu. Technol.*, 15(4), ep464 (2023).
<https://doi.org/10.30935/cedtech/13605>
9. Noy, S., & Zhang, W. Experimental evidence on the productivity effects of generative artificial intelligence. *Sci.*, 381(6654), 187–192 (2023). <https://doi.org/10.1126/science.adh2586>
10. Eloundou, T., Manning, S., Mishkin, P., & Rock, D. (2023). GPTs are GPTs: An early look at the labor market impact potential of large language models. *arXiv*, arXiv:2303.10130 (2023).
<https://doi.org/10.48550/arXiv.2303.10130>
11. Mahapatra, S. Impact of ChatGPT on ESL students' academic writing skills: A mixed methods intervention study. *Smart Learn. Environ.*, 11(1), 9 (2024).
<https://doi.org/10.1186/s40561-024-00295-9>
12. Mohammad Ali, A. An Intervention Study on the Use of Artificial Intelligence in the ESL Classroom: English teacher perspectives on the Effectiveness of ChatGPT for Personalized Language Learning [Unpublished manuscript] (2023).
<https://www.diva-portal.org/smash/get/diva2:1872077/FULLTEXT02.pdf>
13. Bhimavarapu, V. The impact of generative AI on human productivity in creative writing. *J. Stu. Res.*, 12(3), 1-9 (2023). <https://doi.org/10.47611/jsrhs.v12i3.4780>

14. Cummings, R. E., Monroe, S. M., & Watkins, M. Generative AI in first-year writing: An early analysis of affordances, limitations, and a framework for the future. *Comput. Compos.*, 71, 102827 (2024). <https://doi.org/10.1016/j.compcom.2024.102827>
15. Choi, J. H., & Schwarcz, D. AI assistance in legal analysis: An empirical study. *Minn. Leg. Studies*, Research Paper No. 23-22 (2023). <https://doi.org/10.2139/ssrn.4539836>
16. Barrot, J. S. Using ChatGPT for second language writing: Pitfalls and potentials. *Assess. Writ.*, 57, 100745 (2023). <https://doi.org/10.1016/j.asw.2023.100745>
17. Huang, J., & Tan, M. The role of ChatGPT in scientific communication: Writing better scientific review articles. *Am. J. Cancer Res.*, 13(4), 1148–1154 (2023).
18. Sallam, M. ChatGPT utility in healthcare education, research, and practice: Systematic review on the promising perspectives and valid concerns. *Healthc.*, 11(6), 887 (2023).
<https://doi.org/10.3390/healthcare11060887>
19. Al-Zaghir, Z., Matroud, A. A., Abdalgader, K., & Alnaqbi, N. Exploring opportunities and challenges of using ChatGPT in professional writing instruction. *2023 IEEE Intl. Conf. Teach., Assess. Learn. Eng. (TALE)*, 1-6 (2023). <https://doi.org/10.1109/TALE56641.2023.10398234>
20. Wu, Y. Study on the impact of utilizing ChatGPT and other AI tools for feedback in EAP writing classrooms on the discursive writing performance of english major students. *Transact. Soc. Sci., Edu. Hum. Res.*, 4, 143–150 (2024). <https://doi.org/10.62051/4se95x52>
21. Crothers, E. N., Japkowicz, N., & Viktor, H. L. Machine-generated text: A Comprehensive survey of threat models and detection methods. *IEEE Access*, 11, 70977–71002 (2023).
<https://doi.org/10.1109/ACCESS.2023.3294090>
22. Lingard, L. Writing with ChatGPT: An illustration of its capacity, limitations & implications for academic writers. *Perspect. Med. Educ.*, 12(1), 261–270 (2023).
<https://doi.org/10.5334/pme.1072>

23. Chiu, T. K. F. Future research recommendations for transforming higher education with generative AI. *Comput. Educ.: Artif. Intell.*, 6, 100197 (2024).
<https://doi.org/10.1016/j.caeai.2023.100197>
24. Buriak, J. M., Akinwande, D., Artzi, N., Brinker, C. J., Burrows, C., Chan, W. C. W., Chen, C., Chen, X., Chhowalla, M., Chi, L., Chueh, W., Crudden, C. M., Di Carlo, D., Glotzer, S. C., Hersam, M. C., Ho, D., Hu, T. Y., Huang, J., Javey, A., ... Ye, J. Best practices for using AI when writing scientific manuscripts: Caution, care, and consideration: Creative science depends on it. *ACS Nano*, 17(5), 4091–4093 (2023). <https://doi.org/10.1021/acsnano.3c01544>
25. Brynjolfsson, E., Li, D., & Raymond, L. R. (2023). Generative AI at Work. *NBER*, Working paper 31161 (2023). <https://doi.org/10.3386/w31161>
26. Al Naqbi, H., Bahroun, Z., & Ahmed, V. Enhancing work productivity through generative artificial intelligence: A comprehensive literature review. *Sustainability*, 16(3), 1166 (2024).
<https://doi.org/10.3390/su16031166>
27. Garg, R. K., Urs, V. L., Agarwal, A. A., Chaudhary, S. K., Paliwal, V., & Kar, S. K. Exploring the role of ChatGPT in patient care (diagnosis and treatment) and medical research: A systematic review. *Health Promot. Perspect.*, 13(3), 183–191 (2023). <https://doi.org/10.34172/hpp.2023.22>
28. Lazkani, O. Revolutionizing education of art and design through ChatGPT. In A. Al-Marzouqi, S. A. Salloum, M. Al-Saidat, A. Aburayya, & B. Gupta (Eds.), *Artificial Intelligence in Education: The Power and Dangers of ChatGPT in the Classroom* (pp. 49–60). Springer Nature Switzerland (2024). https://doi.org/10.1007/978-3-031-52280-2_4

Supplementary Material 1 - Memo Writing Assignment Version C

Assignment Prompt: You are the Senior Editor of the Pittsburgh Times, an online news website that specializes in reliable and factual national and local news coverage. The CEO sent you the following email:

Hi Senior Editor,

I've been reading about the increasing use of Generative AI tools in newsrooms and came across this article: New survey finds half of newsrooms use Generative AI tools; only 20% have guidelines in place (source: <https://wan-ifra.org/2023/05/new-genai-survey/>). I am struck by the following survey results:

- Nearly half (49%) of newsrooms are using Generative AI tools like ChatGPT.
- 70% of survey participants believe these tools are helpful for their newsrooms

Can you please send me an email with your thoughts, in particular:

- Summarize the article for me
- Find out if the survey result findings I call out above are representative of current opinion (e.g., see if there are other survey results that align or conflict)
- Briefly explain the risks and benefits of GenerativeAI (e.g., ChatGPT) in newsrooms in a short summary
- Recommend a clear and succinct policy for our journalists to use ChatGPT for our newsroom. Feel free to pull in what you think is useful from other news organizations. Our policy should balance the usefulness of the tool with journalistic integrity.

I am a busy CEO, so please keep this no longer than necessary! 2-3 pages will more than suffice.

Thanks,
The CEO

Additional assignment requirements:

- **Citation:** If you find relevant information on the Internet, such as other survey results or other University policies you borrow from, please cite them and include a link in parentheses after the sentence where the material is used.
- **Format:** Please submit to Canvas in a document, but format it as an email.
- **Length:** 2-3 pages single-spaced
- **VERY IMPORTANT:** Please track and calculate the time it takes you to finish the assignment. You should only track active time engaged in the assignment, exclude any breaks you take. The time taken will not reflect at all on your grade. You could use your phone's stopwatch, for example. If you forget to stop or start the timer, please use the best estimate you have.

Supplementary Material 2 - Grading Rubric

	Proficient - 3 points	Sufficient - 2 points	Insufficient - 1 point
Summarization	Well-written, concise summary that hits key points	Summary included, but lacks conciseness or completeness	Insufficient summary
External Survey Results Finding			
External survey research	Included sufficient and relevant external research	Some external research is included, but may not be the best fit	Did not identify enough or relevant external research
Integration and analysis of survey results	Research is well-integrated and analyzed	Research is Integrated, but analysis could be improved	Not integrated or analyzed sufficiently
Policy recommendation			
Articulation of policy	Clear articulation of a policy position	Policy is stated but may not be clear	Lack of a comprehensible policy
Policy is logical and holistic	Policy is logically stated and comprehensive.	Policy is logical but unclear in places or does not account for major issues.	Policy statement is illogical and does not account for most major issues.
Argument and analysis to support policy	Strong and comprehensive logical arguments to support policy	Made arguments to support policy but missed key issues	Insufficient argument to support policy
Writing Style (style, flow, organization)			
Writing style, flow, and concision	Writing is clear, concise and precise, appropriate vocabulary and variety of sentence structure promotes readability	Writing is clear, but occasionally is ambiguous or lacks precision and coherence; basic sentence structures and/or wordiness	Writing is often unclear, wordy, or incoherent, over reliance on simple or repetitive sentence structures, inappropriate vocabulary
Organization	Strong logical structure that is easy to follow, paragraphs have clear ideas and supporting details, overall unity of ideas and sections working together to support the concepts	Ideas are generally organized logically, although some paragraphs or sections may lack structure or transitions,	Lack of organization of ideas and writing lacks unity, with ideas and sections that do not support the central theme or argument

Supplementary Material 3 - Overview of Course Instruction on Generative AI

The course included the following instruction across the first three weeks of classes:

(1) **Basic operation of GenAI applications, using ChatGPT and Microsoft Copilot as primary example tools.** This included a breakdown of the user interface, how inputs lead to outputs, and various other features such as search access, chat history, and model or mode choices. This also included examples of web retrieval to incorporate real-time web knowledge. Teaching techniques included pre-reading and live demonstrations where students followed along on their own computers.

(2) **Development of GenAI models and applications.** This covered what data models are trained upon to help students understand what the models may know. This also covered language encoding, basics of deep learning, and how models learn to predict the next word through pre-training, understanding mechanisms (e.g., attention), and word embedding. This further covered how models are trained as assistants via fine-tuning and human feedback mechanisms. This included an understanding of the role of meta-prompts, how search retrieval operates in the context of a conversation, and the role of chat session history. Teaching techniques included pre-reading, videos, and in-class lectures, discussions, and live demonstrations where students followed along on their own computers.

(3) **Prompt engineering basics.** This informed students on how prompts are interpreted by genAI applications and how to build context and ask for structure to improve genAI utility. Techniques included explaining in the prompt the goals of the interaction, the role the users seeks the genAI application to play, and how providing deeper context can help improve performance. Instruction covered zero shot, one shot, and few shot prompting, and the tree of thought

approach. Instruction also showed students examples of hallucinations and how to prevent hallucinations by avoiding common failure modes where the model is unlikely to know the answer. Teaching techniques included live in-class lectures and demonstrations where students followed along on their own computers.

(4) **Challenges of overreliance on GenAI.** This taught students the limitations and risks of using GenAI without proper verification and critical thinking. The teaching technique used was a simulation of a time-pressured knowledge work task where the deliverable is extremely difficult to complete in fullness and with complete accuracy (full genAI assignment #2). Thus, students were tasked with a graded research and writing assignment with extremely limited time and instructed to use GenAI as much as possible. This exposed students to outputs that are incorrect, incomplete, biased, or incompatible with the facts they need for their arguments. By seeing these results themselves, students learned to check the sources and quality of the genAI output and also learn where GenAI succeeds and fails.

(5) **Using GenAI as a writing assistant:** This showed students via live demonstrations the following ways to use GenAI as part of the writing process: setting proper context, learning about topics, web research retrieval, information synthesis across sources, summarization across sources, writing organization, writing improvement (e.g., condense, expand, clarify, analyze, explain), developing arguments, idea generation, and providing critical constructive feedback. Moreover, these demonstrations revealed some of the limitations of GenAI such as the tendency to write long, boring responses. Students were warned about depending too much on the tool and the need to check facts and citations.